SECP2523

DATABASE (WBL)

SECTION 02

ASSIGNMENT 1

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QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	Order	Customer
Information	iPhone 16	John

QUESTION 2

Answer the questions below based on the following relational schema:

can only be primary key for one entity

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)
EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)
PROJECT (Proj_Code, Proj_Name, Proj_StartDate)
SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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usually the first is primary key

a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M) comp generated & cannot be noun

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	–
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	–
SCIENTIST_EXPERIMENT	Time	Sc_ID , Exp_NO

one time one experiment

b) Does the table exhibit entity integrity? Answer **'Yes'** or **'No'** and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	have primary key
EXPERIMENT	Yes	have primary key
PROJECT	Yes	have primary key

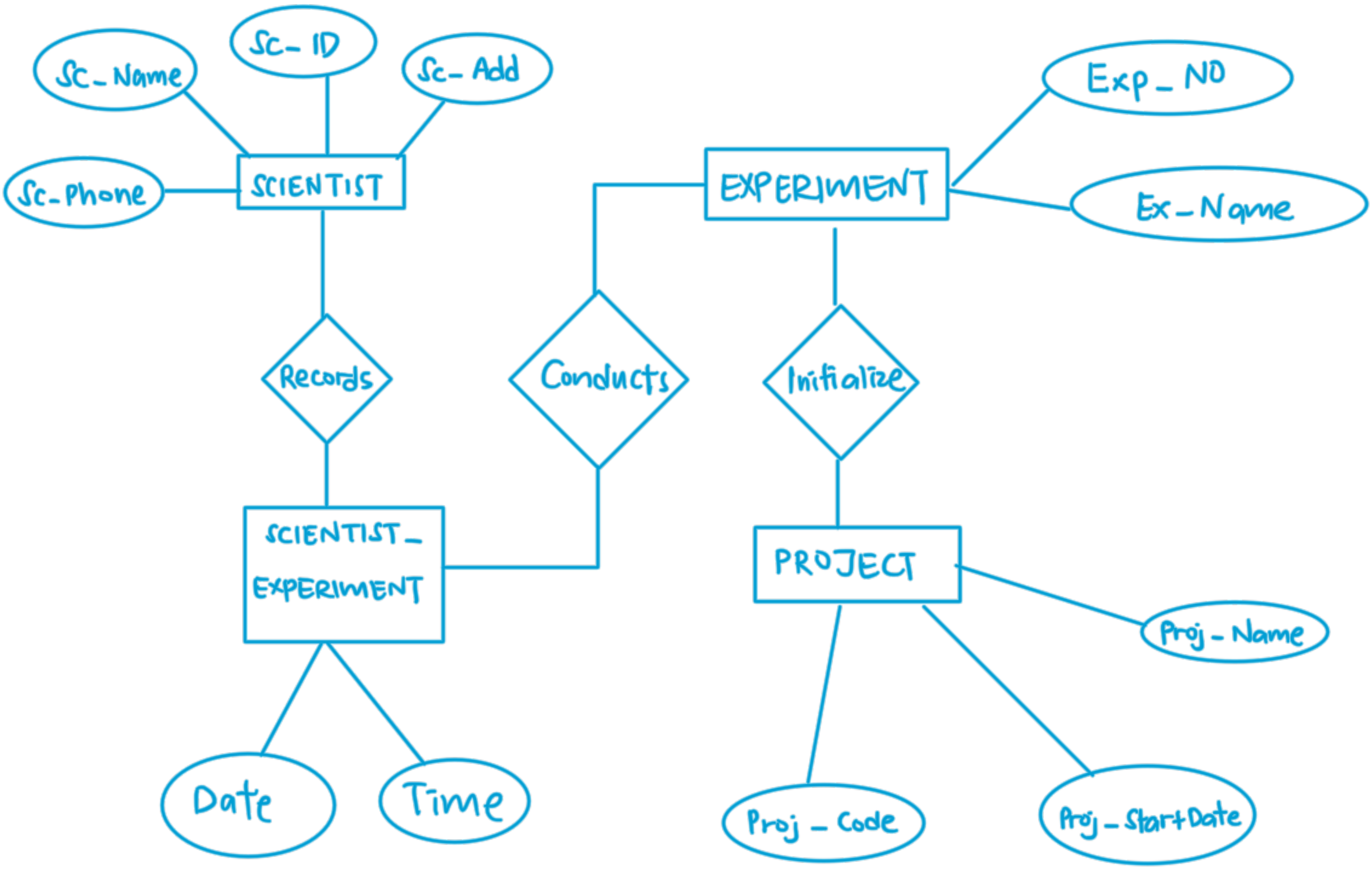
SCIENTIST_EXPERIMENT	Yes	have primary key

c) Does the table exhibit referential integrity? Answer **'Yes'** or '**No'** and then explain your answer. (4M)

entity have foreign key

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	entity do not have foreign key
EXPERIMENT	Yes	entity have foreign key
PROJECT	No	entity do not have foreign key
SCIENTIST_EXPERIMENT	Yes	entity have foreign key

d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

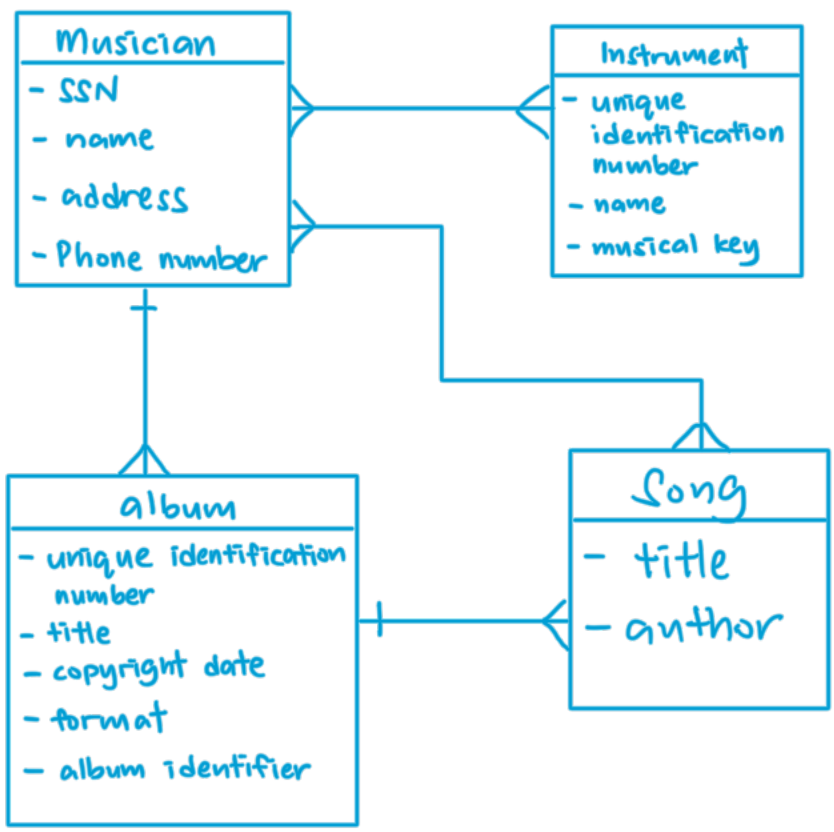
Consider the following about Luxury Records:

Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

Task:

1. List each Entity that is identified in Luxury Records System.
2. For each entity list the attributes.
3. Draw a logical model to represent each entity and the relationships between them.
4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.



QUESTION 4

Give an Give an example of each of the three types of relationships.

i) One-to-many relationship

one people buy many car

ii) Many-to-many relationship

many customers buy many products

iii) One-to-one relationship

one student study one program

iv) Many-to-one relationship

many car belongs to one person

QUESTION 5

CLIENT (Client_Number, Clent_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number) for. key pri. key **RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)**

a) What is the relationship between Recruiter and Client?

many to one (many recruiter to one client)

b) What is the primary key for the Client table and Recruiter table?

Client_Number , Recruiter_Number

c) Define foreign key? Identify foreign key in the Client table.

Foreign key is a set of attributes in a table that refers to the primary key of another table.

Foreign key : Recruiter_Number

QUESTION 6

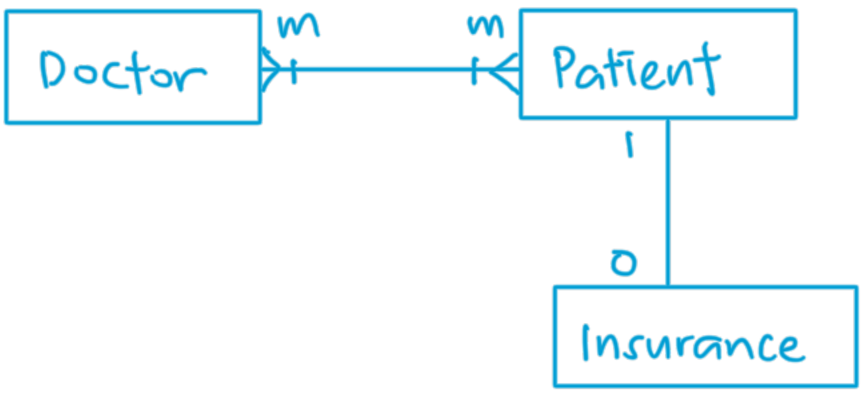
Draw the Chen & Crow's Foot notation based on the following statements:

i) "A customer can make many payments, but each payment is made by only one customer."

ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan." choose many to do

i)

ii)



many

one

QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book_ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.